Machine Learning (N-PECCS602P)

S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 7

Aim: To apply Support Vector Machine for classification and analyze the impact of different kernels on model accuracy.

Name of Student : _____

Roll No. : _____

Semester/Year : _____

Academic Session : _____

Date of Performance : _____

Date of Submission : _____

Department of Computer Science & Engineering, S.B.J.I.T.M.R, Nagpur.

Aim: To apply Support Vector Machine for classification and analyze the impact of different kernels on model accuracy.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

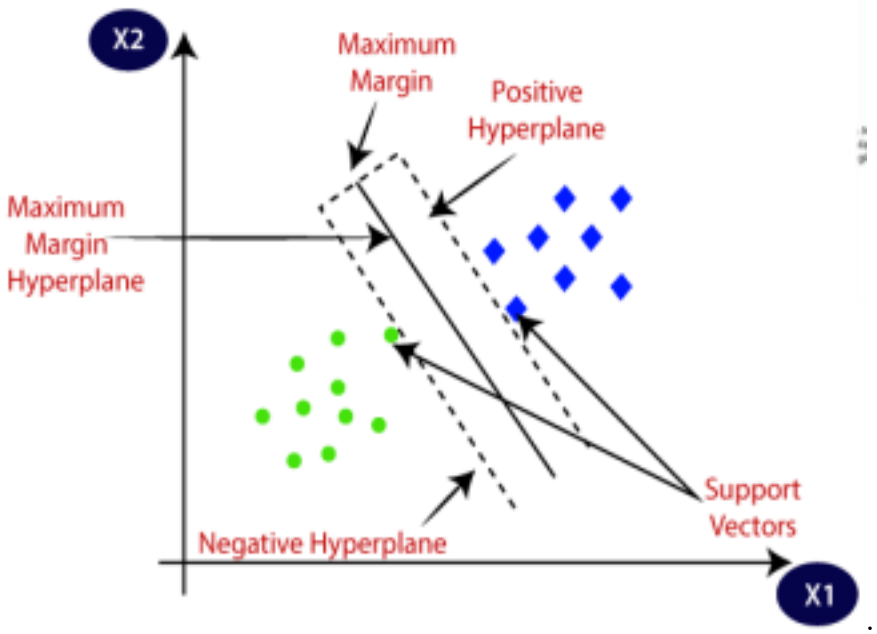
- The objective of the SVM algorithm is to find a hyperplane in an N-dimensional space that distinctly classifies the data points.
- The dimension of the hyperplane depends upon the number of features. ● If the number of input features is two, then the hyperplane is just a line. ● If the number of input features is three, then the hyperplane becomes a 2-D plane. It becomes difficult to imagine when the number of features exceeds three.

THEORY:

Support Vector Machine or SVM is one of the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems. However, primarily, it is used for Classification problems in Machine Learning.

The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future. This best decision boundary is called a hyperplane.

SVM chooses the extreme points/vectors that help in creating the hyperplane. These extreme cases are called as support vectors, and hence algorithm is termed as Support Vector Machine. Consider the below diagram in which there are two different categories that are classified using a decision boundary or hyperplane:



Types of SVM

SVM can be of two types:

- **Linear SVM:** Linear SVM is used for linearly separable data, which means if a dataset can be classified into two classes by using a single straight line, then such data is termed as linearly separable data, and classifier is used called as Linear SVM classifier.
- **Non-linear SVM:** Non-Linear SVM is used for non-linearly separated data, which means if a dataset cannot be classified by using a straight line, then such data is termed as non-linear data and classifier used is called as Non-linear SVM classifier.

Hyperplane and Support Vectors in the SVM algorithm:

Hyperplane: There can be multiple lines/decision boundaries to segregate the classes in n dimensional space, but we need to find out the best decision boundary that helps to classify the data points. This best boundary is known as the hyperplane of SVM.

The dimensions of the hyperplane depend on the features present in the dataset, which means if there are 2 features (as shown in image), then hyperplane will be a straight line. And if there are 3 features, then hyperplane will be a 2-dimension plane.

We always create a hyperplane that has a maximum margin, which means the maximum distance between the data points.

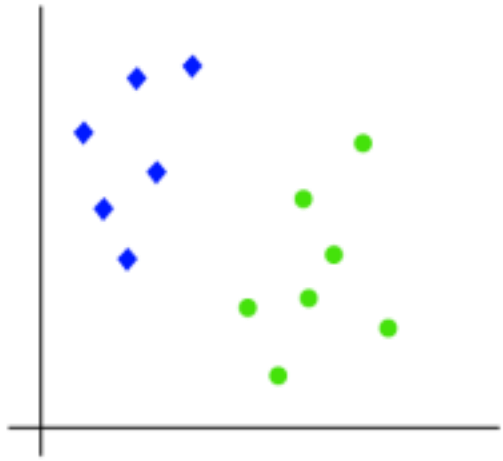
Support Vectors:

The data points or vectors that are the closest to the hyperplane and which affect the position of the hyperplane are termed as Support Vector. Since these vectors support the hyperplane, hence called a Support vector.

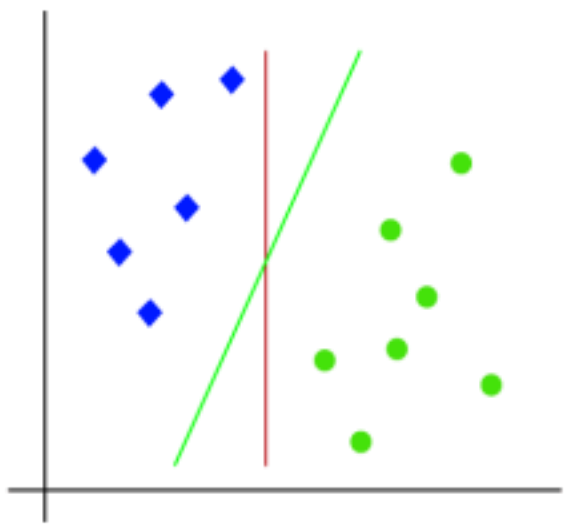
How does SVM works?

Linear SVM:

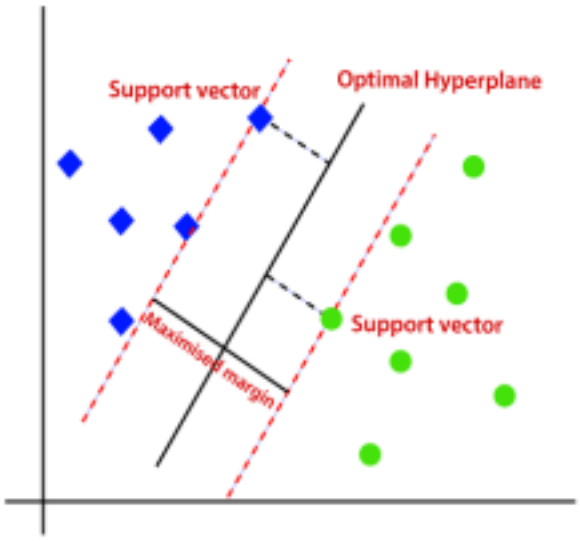
The working of the SVM algorithm can be understood by using an example. Suppose we have a dataset that has two tags (green and blue), and the dataset has two features x1 and x2. We want a classifier that can classify the pair(x1, x2) of coordinates in either green or blue. Consider the below image:



So as it is 2-d space so by just using a straight line, we can easily separate these two classes. But there can be multiple lines that can separate these classes. Consider the below image:

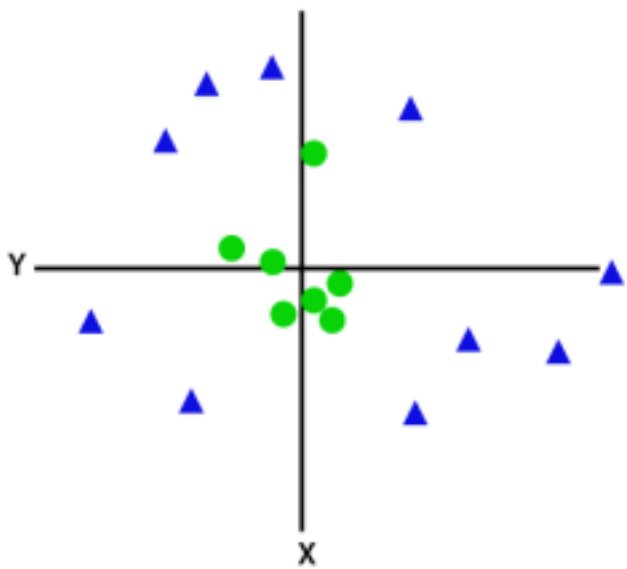


Hence, the SVM algorithm helps to find the best line or decision boundary; this best boundary or region is called as a **hyperplane**. SVM algorithm finds the closest point of the lines from both the classes. These points are called support vectors. The distance between the vectors and the hyperplane is called as **margin**. And the goal of SVM is to maximize this margin. The **hyperplane** with maximum margin is called the **optimal hyperplane**.



Non-Linear SVM:

If data is linearly arranged, then we can separate it by using a straight line, but for non-linear data, we cannot draw a single straight line. Consider the below image:



So to separate these data points, we need to add one more dimension. For linear data, we have used two dimensions x and y, so for non-linear data, we will add a third dimension z. It can be calculated as:

$z=x^2+y^2$

By adding the third dimension, the sample space will become as below

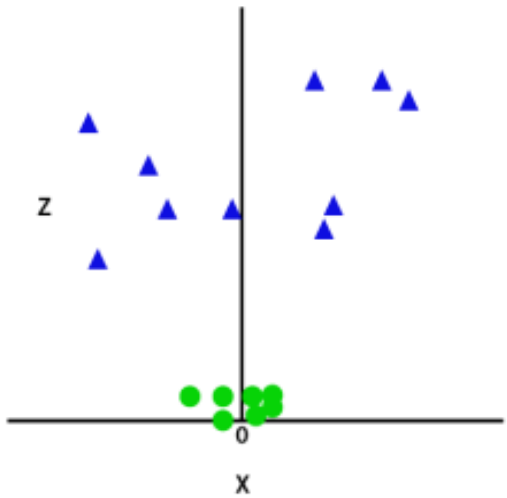
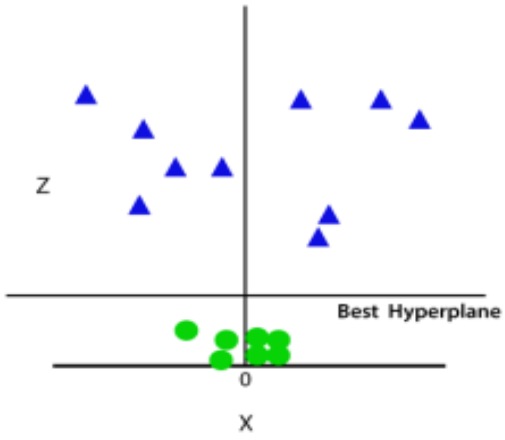
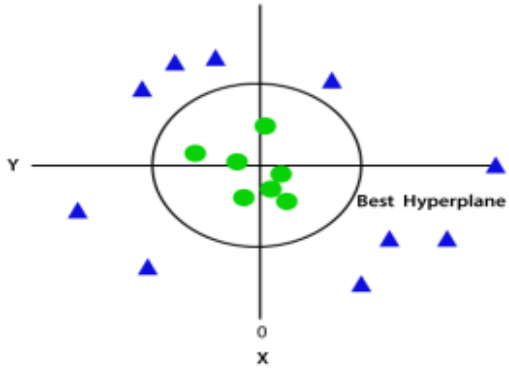


image:

So now, SVM will divide the datasets into classes in the following way. Consider the below image:



Since we are in 3-d Space, hence it is looking like a plane parallel to the x-axis. If we convert it in 2d space with z=1, then it will become as:



Hence we get a circumference of radius 1 in case of non-linear data.

Advantages of SVM

1. SVM works better when the data is Linear
2. It is more effective in high dimensions
3. With the help of the kernel trick, we can solve any complex problem
4. SVM is not sensitive to outliers
5. Can help us with Image classification

Disadvantages of SVM

1. Choosing a good kernel is not easy
2. It doesn't show good results on a big dataset
3. The SVM hyperparameters are Cost -C and gamma. It is not that easy to fine-tune these hyper parameters. It is hard to visualize their impact

PROGRAM CODE:

OUTPUT (SCREENSHOT):

MLpract7.ipynb Cannot save changes

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all Copy to Drive Connect Share

Prathamesh Deshmukh (CS23115)

Ml Lab-7

Aim:- To apply support Vector Machine for classification analyze the impact of different Kernels on model accuracy

```
import pandas as pd
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

```
from google.colab import drive
drive.mount("/content/drive")
```

```
Mounted at /content/drive
```

```
path= "/content/drive/MyDrive/ML_Dataset/pulsar_stars.csv"
df=pd.read_csv(path)
df.head(5)
```

	Mean of the integrated profile	Standard deviation of the integrated profile	Excess kurtosis of the integrated profile	Skewness of the integrated profile	Mean of the DM-SNR curve	Standard deviation of the DM-SNR curve	Excess kurtosis of the DM-SNR curve	Skewness of the DM-SNR curve	target_class
0	140.562500	55.683782	-0.234571	-0.699648	3.199833	19.110426	7.975532	74.242225	0
1	102.507812	58.882430	0.465318	-0.515088	1.677258	14.860146	10.576487	127.393580	0
2	103.015625	39.341649	0.323328	1.051164	3.121237	21.744669	7.735822	63.171909	0
3	136.750000	57.178449	-0.068415	-0.636238	3.642977	20.959280	6.896499	53.593661	0
4	88.726562	40.672225	0.600866	1.123492	1.178930	11.468720	14.269573	252.567306	0

```
df.shape
```

```
(17898, 9)
```

```python
df.describe()
```

	Mean of the integrated profile	Standard deviation of the integrated profile	Excess kurtosis of the integrated profile	Skewness of the integrated profile	Mean of the DM-SNR curve	Standard deviation of the DM-SNR curve	Excess kurtosis of the DM-SNR curve	Skewness of the DM-SNR curve	target_class
count	17898.000000	17898.000000	17898.000000	17898.000000	17898.000000	17898.000000	17898.000000	17898.000000	17898.000000
mean	111.079968	46.549532	0.477857	1.770279	12.614400	26.326515	8.303556	104.857709	0.091574
std	25.652935	6.843189	1.064040	6.167913	29.472897	19.470572	4.506092	106.514540	0.288432
min	5.812500	24.772042	-1.876011	-1.791886	0.213211	7.370432	-3.139270	-1.976976	0.000000
25%	100.929688	42.376018	0.027098	-0.188572	1.923077	14.437332	5.781506	34.960504	0.000000
50%	115.078125	46.947479	0.223240	0.198710	2.801839	18.461316	8.433515	83.064556	0.000000
75%	127.085938	51.023202	0.473325	0.927783	5.464256	28.428104	10.702959	139.309330	0.000000
max	192.617188	98.778911	8.069522	68.101622	223.392141	110.642211	34.539844	1191.000837	1.000000

```python
# Remove leading and trailing spaces from column names
df.columns = df.columns.str.strip()
print(df.columns)
```

```
Index(['Mean of the integrated profile',
       'Standard deviation of the integrated profile',
       'Excess kurtosis of the integrated profile',
       'Skewness of the integrated profile', 'Mean of the DM-SNR curve',
       'Standard deviation of the DM-SNR curve',
       'Excess kurtosis of the DM-SNR curve', 'Skewness of the DM-SNR curve',
       'target_class'],
      dtype='object')
```


```python
# Remove leading and trailing spaces from column names
df.columns = df.columns.str.strip()
print(df.columns)
```

```
Index(['Mean of the integrated profile',
       'Standard deviation of the integrated profile',
       'Excess kurtosis of the integrated profile',
       'Skewness of the integrated profile', 'Mean of the DM-SNR curve',
       'Standard deviation of the DM-SNR curve',
       'Excess kurtosis of the DM-SNR curve', 'Skewness of the DM-SNR curve',
       'target_class'],
      dtype='object')
```


```python
df = df.rename(columns={
    'Mean of the integrated profile': 'MIP',
    'Standard deviation of the integrated profile': 'SDIP',
    'Excess kurtosis of the integrated profile': 'EKIP',
    'Skewness of the integrated profile': 'SKIP',
    'Mean of the DM-SNR curve': 'MDSNR',
    'Standard deviation of the DM-SNR curve': 'SDSNR',
    'Excess kurtosis of the DM-SNR curve': 'EKSNR',
    'Skewness of the DM-SNR curve': 'SKSNR',
    'target_class': 'Target'
})

print(df.columns)
```

```
Index(['MIP', 'SDIP', 'EKIP', 'SKIP', 'MDSNR', 'SDSNR', 'EKSNR', 'SKSNR',
       'Target'],
      dtype='object')
```

```python
df.describe()
```

	MIP	SDIP	EKIP	SKIP	MDSNR	SDSNR	EKSNR	SKSNR	Target
count	17898.000000	17898.000000	17898.000000	17898.000000	17898.000000	17898.000000	17898.000000	17898.000000	17898.000000
mean	111.079968	46.549532	0.477857	1.770279	12.614400	26.326515	8.303556	104.857709	0.091574
std	25.652935	6.843189	1.064040	6.167913	29.472897	19.470572	4.506092	106.514540	0.288432
min	5.812500	24.772042	-1.876011	-1.791886	0.213211	7.370432	-3.139270	-1.976976	0.000000
25%	100.929688	42.376018	0.027098	-0.188572	1.923077	14.437332	5.781506	34.960504	0.000000
50%	115.078125	46.947479	0.223240	0.198710	2.801839	18.461316	8.433515	83.064556	0.000000
75%	127.085938	51.023202	0.473325	0.927783	5.464256	28.428104	10.702959	139.309330	0.000000
max	192.617188	98.778911	8.069522	68.101622	223.392141	110.642211	34.539844	1191.000837	1.000000

```python
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
```

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

```
df.columns
```

```
Index(['MIP', 'SDIP', 'EKIP', 'SKIP', 'MDSNR', 'SDSNR', 'EKSNR', 'SKSNR',
       'Target'],
      dtype='object')
```

```
print(df['Target'].value_counts())
```

```
Target
0    16259
1     1639
Name: count, dtype: int64
```

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
X_train
```

Show hidden output

```
X_test
```

Show hidden output

```
X
```

	MIP	SDIP	EKIP	SKIP	MDSNR	SDSNR	EKSNR	SKSNR
0	140.562500	55.683782	-0.234571	-0.699648	3.199833	19.110426	7.975532	74.242225
1	102.507812	58.882430	0.465318	-0.515088	1.677258	14.860146	10.576487	127.393580
2	103.015625	39.341649	0.323328	1.051164	3.121237	21.744669	7.735822	63.171909
3	136.750000	57.178449	-0.068415	-0.636238	3.642977	20.959280	6.896499	53.593661
4	88.726562	40.672225	0.600866	1.123492	1.178930	11.468720	14.269573	252.567306
...	...	...	...	...	...	...	...	...
17893	136.429688	59.847421	-0.187846	-0.738123	1.296823	12.166062	15.450260	285.931022
17894	122.554688	49.485605	0.127978	0.323061	16.409699	44.626893	2.945244	8.297092
17895	119.335938	59.935939	0.159363	-0.743025	21.430602	58.872000	2.499517	4.595173
17896	114.507812	53.902400	0.201161	-0.024789	1.946488	13.381731	10.007967	134.238910
17897	57.062500	85.797340	1.406391	0.089520	188.306020	64.712562	-1.597527	1.429475

17898 rows × 8 columns

Vector Machine

```
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score
svc=SVC()
svc.fit(X_train,y_train)
y_pred=svc.predict(X_test)
accuracy=accuracy_score(y_test,y_pred)
print('Model accuracy score with default hyperparameters: {0:0.4f}'. format(accuracy))
```

```
Model accuracy score with default hyperparameters: 0.9790
```

```
from sklearn.metrics import confusion_matrix, accuracy_score, classification_report
```

```
print('Confusion Matrix')
cm = confusion_matrix(y_pred,y_test)
print(confusion_matrix(y_pred,y_test))
```

```
Confusion Matrix
[[4863   92]
 [  21  394]]
```

```
print('Accuracy Score')
print(accuracy_score(y_pred,y_test))
```

```
Accuracy Score
0.9789571694599628
```

```
print('Classification Report')
print(classification_report(y_pred,y_test))
```

```
Classification Report
              precision    recall  f1-score   support

           0       1.00      0.98      0.99      4955
           1       0.81      0.95      0.87       415

    accuracy                           0.98      5370
   macro avg       0.90      0.97      0.93      5370
weighted avg       0.98      0.98      0.98      5370
```

```
svc=SVC(C=1.0)
svc.fit(X_train,y_train)
y_pred=svc.predict(X_test)
print('Model accuracy score with rbf kernel and c=1.0 : {0:.4f}'.format(accuracy))
```

```
Model accuracy score with rbf kernel and c=1.0 : 0.9790
```

+ Code + Text

```
svc=SVC(C=100.0)
svc.fit(X_train,y_train)
y_pred=svc.predict(X_test)
print('Model accuracy score with rbf kernel and c=100.0 : {0:.4f}'.format(accuracy))
```

```
Model accuracy score with rbf kernel and c=100.0 : 0.9790
```

```
svc=SVC(C=1000.0)
svc.fit(X_train,y_train)
y_pred=svc.predict(X_test)
print('Model accuracy score with rbf kernel and c=100.0 : {0:.4f}'.format(accuracy))
```

```
Model accuracy score with rbf kernel and c=100.0 : 0.9790
```

SVM with linear

```
linear_svc=SVC(kernel='linear',C=1.0)
linear_svc.fit(X_train,y_train)
y_pred_test=linear_svc.predict(X_test)
linear_accuracy = accuracy_score(y_test,y_pred_test)
print('Model accuracy score with linear kernel and c=1.0 : {0:.4f}'.format(linear_accuracy))
```

```
Model accuracy score with linear kernel and c=1.0 : 0.9788
```

```
linear_svc=SVC(kernel='linear',C=100.0)
linear_svc.fit(X_train,y_train)
y_pred_test=linear_svc.predict(X_test)
linear_accuracy = accuracy_score(y_test,y_pred_test)
print('Model accuracy score with linear kernel and c=100.0 : {0:.4f}'.format(linear_accuracy))
```

```
Model accuracy score with linear kernel and c=1.0 : 0.9786
```

+ Code + Text

```
linear_svc=SVC(kernel='linear',C=1000.0)
linear_svc.fit(X_train,y_train)
y_pred_test=linear_svc.predict(X_test)
linear_accuracy = accuracy_score(y_test,y_pred_test)
print('Model accuracy score with linear kernel and c=1000.0 : {0:.4f}'.format(linear_accuracy))
```

```
Model accuracy score with linear kernel and c=1000.0 : 0.9786
```

SVM with polynomial

```
poly_svc=SVC(kernel='poly',C=1.0)
poly_svc.fit(X_train,y_train)
y_pred_test=poly_svc.predict(X_test)
poly_accuracy = accuracy_score(y_test,y_pred_test)
print('Model accuracy score with polynomial kernel and c=1.0 : {0:.4f}'.format(poly_accuracy))
```

```
Model accuracy score with polynomial kernel and c=1.0 : 0.9769
```

```
poly_svc=SVC(kernel='poly',C=100.0)
poly_svc.fit(X_train,y_train)
y_pred_test=poly_svc.predict(X_test)
poly_accuracy = accuracy_score(y_test,y_pred_test)
print('Model accuracy score with polynomial kernel and c=100.0 : {0:.4f}'.format(poly_accuracy))
```

```
Model accuracy score with polynomial kernel and c=100.0 : 0.9793
```

```
poly_svc=SVC(kernel='poly',C=1000.0)
poly_svc.fit(X_train,y_train)
y_pred_test=poly_svc.predict(X_test)
poly_accuracy = accuracy_score(y_test,y_pred_test)
print('Model accuracy score with polynomial kernel and c=1000.0 : {0:.4f}'.format(poly_accuracy))
```

```
Model accuracy score with polynomial kernel and c=1000.0 : 0.9793
```

svm with sigmoid

```
[ ]
sigmoid_svc=SVC(kernel='sigmoid',C=1.0)
sigmoid_svc.fit(X_train,y_train)
y_pred_test=sigmoid_svc.predict(X_test)
sigmoid_accuracy = accuracy_score(y_test,y_pred_test)
print('Model accuracy score with sigmoid kernel and c=1.0 : {0:.4f}'.format(sigmoid_accuracy))

Model accuracy score with sigmoid kernel and c=1.0 : 0.8773
```

```
[ ]
sigmoid_svc=SVC(kernel='sigmoid',C=100.0)
sigmoid_svc.fit(X_train,y_train)
y_pred_test=sigmoid_svc.predict(X_test)
sigmoid_accuracy = accuracy_score(y_test,y_pred_test)
print('Model accuracy score with sigmoid kernel and c=100.0 : {0:.4f}'.format(sigmoid_accuracy))

Model accuracy score with sigmoid kernel and c=100.0 : 0.8773
```

```
[ ]
sigmoid_svc=SVC(kernel='sigmoid',C=1000.0)
sigmoid_svc.fit(X_train,y_train)
y_pred_test=sigmoid_svc.predict(X_test)
sigmoid_accuracy = accuracy_score(y_test,y_pred_test)
print('Model accuracy score with sigmoid kernel and c=1000.0 : {0:.4f}'.format(sigmoid_accuracy))

Model accuracy score with sigmoid kernel and c=1000.0 : 0.8773
```

```
[ ]
print('Confusion Matrix')
cm = confusion_matrix(y_pred,y_test)
print(confusion_matrix(y_pred,y_test))

Confusion Matrix
[[4863   92]
 [  21  394]]
```

CONCLUSION:

_____

_____

_____

_____

DISCUSSION AND VIVA VOCE:

1. In a real-world fraud detection system, when would you prefer SVM over Logistic Regression?

Ans :-

_____

_____

_____

2. How does the choice of kernel (linear, RBF, polynomial) impact performance in production systems?

Ans :-

_____

_____

3. What is the role of the regularization parameter C in SVM, and how does it affect overfitting?

Ans :-

4. In high-dimensional datasets like text classification, why is linear SVM often preferred?

Ans :-

5. How would you handle imbalanced data when training an SVM model in a medical diagnosis application?

Ans :-

6. Why is feature scaling mandatory before applying SVM?

Ans :-

7. What are the computational challenges of deploying SVM in large-scale industry datasets?

Ans :-

8. In a real-time prediction system (e.g., credit scoring), what are the limitations of SVM?

Ans :-

9. How do you select hyperparameters (C and gamma) in industry projects?

Ans :-

10. Why might companies prefer tree-based models (Random Forest, XGBoost) over SVM in production?

Ans :-

REFERENCE

- https://www.analyticsvidhya.com/blog/2021/10/support-vector-machinessvm-a complete-guide-for-beginners/
- https://www.geeksforgeeks.org/support-vector-machine-algorithm/
- https://www.javatpoint.com/machine-learning-support-vector-machine-algorithm

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